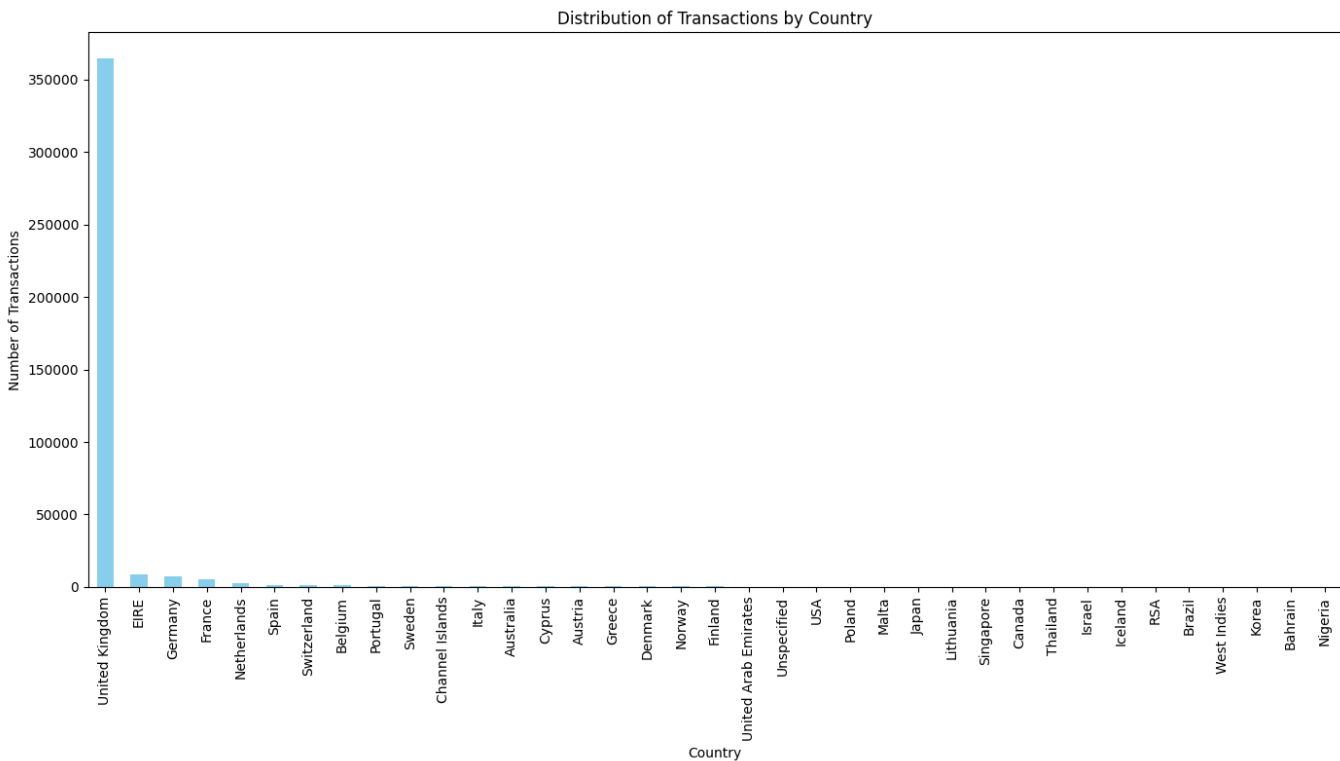


Comprehensive Report on Online Retail II Dataset

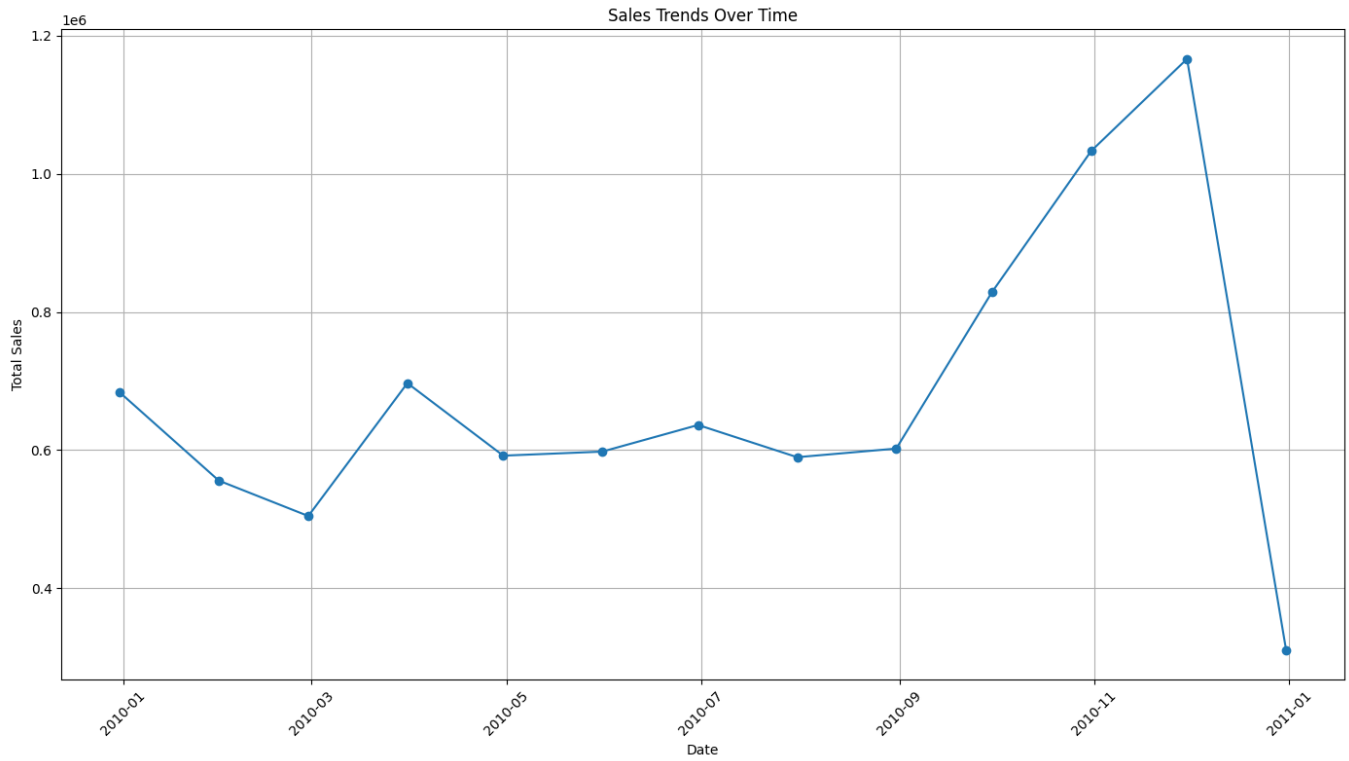
1. Data Visualization

1.1 Distribution of Transactions by Country



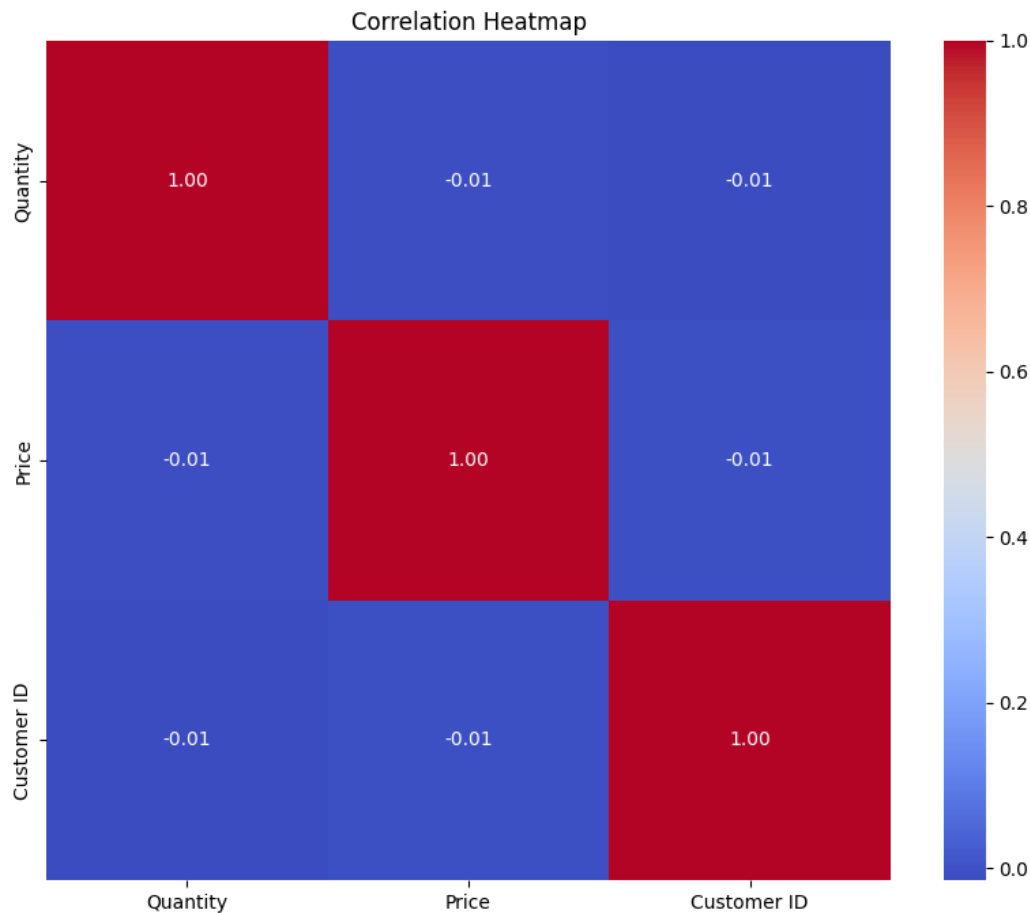
The bar chart above shows the distribution of transactions by country. It highlights the countries with the highest number of transactions, indicating potential key markets for the business.

1.2 Sales Trends Over Time



The time series plot illustrates the sales trends over the specified period. It helps in identifying seasonal patterns and sales peaks, which can be crucial for inventory and marketing strategies.

1.3 Correlation Heatmap



The correlation heatmap provides insights into the relationships between different features. Understanding these correlations can help in feature selection and improving model performance.

2. Data Mining

2.1 Frequent Itemsets

Itemset: frozenset({'WHITE CHERRY LIGHTS'}), Support: 0.0112

Itemset: frozenset({'10 COLOUR SPACEBOY PEN'}), Support: 0.0129

Itemset: frozenset({'12 PENCIL SMALL TUBE WOODLAND'}), Support: 0.0131

Itemset: frozenset({'12 PENCILS SMALL TUBE RED SPOTTY'}), Support: 0.0158

Itemset: frozenset({'12 PENCILS SMALL TUBE SKULL'}), Support: 0.0214

The frequent itemsets identified using the Apriori algorithm reveal combinations of products that are often purchased together. This information can be used to optimize product placement and cross-selling strategies.

2.2 Association Rules

Rule: frozenset({'60 TEATIME FAIRY CAKE CASES'}) -> frozenset({'72 SWEETHEART FAIRY CAKE CASES'}), Confidence: 0.

Rule: frozenset({'72 SWEETHEART FAIRY CAKE CASES'}) -> frozenset({'60 TEATIME FAIRY CAKE CASES'}), Confidence: 0.

Rule: frozenset({'PACK OF 60 DINOSAUR CAKE CASES'}) -> frozenset({'60 TEATIME FAIRY CAKE CASES'}), Confidence: 0.

Rule: frozenset({'60 TEATIME FAIRY CAKE CASES'}) -> frozenset({'PACK OF 60 DINOSAUR CAKE CASES'}), Confidence: 0.

Rule: frozenset({'PACK OF 60 MUSHROOM CAKE CASES'}) -> frozenset({'60 TEATIME FAIRY CAKE CASES'}), Confidence: 0.

The association rules provide insights into the likelihood of purchasing certain products together. High confidence and lift values indicate strong associations, which can inform promotional strategies.

2.3 Popular Products

Product: WHITE HANGING HEART T-LIGHT HOLDER, Frequency: 3107

Product: REGENCY CAKESTAND 3 TIER, Frequency: 1695

Product: STRAWBERRY CERAMIC TRINKET BOX, Frequency: 1372

Product: ASSORTED COLOUR BIRD ORNAMENT, Frequency: 1355

Product: HOME BUILDING BLOCK WORD, Frequency: 1192

The analysis of popular products highlights the top-selling items, which can guide inventory management and marketing focus.